STR	STREAM DESCRIPTION	SOURCE	DESTINATION	TOTAL (kg/h)	SOLIDS (kg/h)	WATER (kg/h)	MOIST (%wb)	T (degC)	P (bar g)	PHASE STATE	VOL FLOW (m3/h)
01	Raw Split Pigeon Pea Infeed	Silo TK-001	Elevator CV-101	937.50	839.06	98.44	10.50%	28.0	ATM	Solid Granular	1.17
02	Treated Soft Process Water	Water Plant WTP-01	Soak Battery TK-101	1,406.25	0.00	1,406.25	100.00%	30.0	2.00	Liquid	1.41
03	Reagent Dosing Solution (10%)	Dosing Skid SK-100	Soak Battery TK-101	14.06	1.41	12.65	90.00%	28.0	3.00	Aqueous Soln	0.01
04	Hydrated Dal Slurry Discharge	Soak Battery TK-101	Screen SC-101	2,357.81	840.47	1,517.34	64.35%	30.0	0.20	Solid-Liquid	2.14
05	Separated Spent Soak Liquor	Screen SC-101	Equalization TK-104	1,018.75	6.44	1,012.31	99.37%	30.0	ATM	Turbid Liquid	1.02
06	Moist Dewatered Dal to Dryer	Screen SC-101	Fluid Bed DR-101	1,339.06	834.03	505.03	37.72%	30.0	ATM	Wet Particulate	1.58
07	Hot Conditioned Drying Air	Air Heater HE-101	Plenum DR-101	14,150.00	0.00	155.65	1.10% db	85.0	0.03	Gas	11,800.0
08	Moist Vapour Exhaust Air	Dryer Hood DR-101	Cyclone CY-101	14,551.31	10.60	556.96	3.83% db	62.0	-0.01	Dusty Gas	12,950.0
09	Finished Tender Toor Dal	Dryer DR-101	Packaging PK-101	926.90	834.21	92.69	10.00%	35.0	ATM	Solid Granular	1.16
10	Evaporator Clean Condensate	Condenser HE-103	Water Tank TK-105	980.00	0.00	980.00	100.00%	45.0	ATM	Distillate Liq	0.99
11	ZLD Concentrate Effluent	3rd Effect EV-101C	Solar Pan / Bagging	38.75	15.50	23.25	60.00%	55.0	0.50	Heavy Slurry	0.03
12	Cyclone Separated Fines	Cyclone CY-101	Packaging Recycle	10.60	9.54	1.06	10.00%	58.0	ATM	Fine Powder	0.02

MASS BALANCE RECONCILIATION & SOLIDS CONSERVATION AUDIT (ASPEN PLUS V14)

FLOWSTREAM / BASIS	SOLIDS (kg/h)	WATER (kg/h)	RECONCILIATION & CLOSURE
01 Raw Grain Infeed	839.06	98.44	100.000% Grain Basis (10.50% wb)
02 Soft Process Water	0.00	1,406.25	Treated Soft Water Makeup
03 Conditioning Reagent	1.41	12.65	Active Mineral Salt Hydration
04 Hydrated Dal Slurry	840.47	1,517.34	Soak Battery Tank Discharge
05 Separated Spent Liquor	6.26	1,012.31	0.75% Organic Leaching Loss
06 Moist Dal to Dryer	834.21	505.03	Dewatered Screen Feed (37.72% wb)
09 Finished Tender Dal	834.21	92.69	99.42% Saleable Product Yield
10 ZLD Clean Condensate	0.00	980.00	96.20% Closed-Loop Distillate Reused
11 Heavy Concentrate Bleed	15.50	23.25	Solar Evaporation / Mineral Bagging
12 Cyclone Separated Fines	9.54	1.06	1.14% Recycled to Product Line
OVERALL BALANCE CLOSURE	840.47 in = 840.47	Net Evap: 412.34 kg/h	EXACT 0.0000% CONVERGENCE ERROR

THERMODYNAMIC AUDIT: Converged in Aspen Plus V14 with NRTL & STEAM-TA property packages.

SOLIDS CONSERVATION: Input (839.06 + 1.41 = 840.47 kg/h) == Output (834.21 + 6.26 = 840.47 kg/h).

EFFLUENT ZERO DISCHARGE: Spent liquor (1,018.75 kg/h) -> 980.00 kg/h recycled distillate (96.20%).

GOVERNANCE AUDIT: Lead Engineer: Sourish Senapati | Academic Oversight: Dr. Archisman Ray (IIT KGP)

THERMAL ENERGY, STEAM CONSUMPTION & ELECTRICAL POWER SCHEDULE

UTILITY CONSUMER	TAG	DUTY / RATING	OPERATING RATE	SUPPLY SPECIFICATION
Dryer Steam Air Heater	HE-101	512.4 kWth	864.5 kg/h Steam	3.0 bar g Saturated (143.6 C)
ZLD Evaporator 1st Effect	EV-101A	285.6 kWth	481.8 kg/h Steam	3.0 bar g Saturated (143.6 C)
Total Plant Motive Steam	BLR-01	798.0 kWth	1,346.3 kg/h Steam	Packaged Dual-Fuel (2.0 TPH)
Primary Fluidizing Blower	BL-101	18.5 kW	14.8 kW Running	415V / 3-Ph / 50Hz (VFD)
Evaporator Calandria Pumps	P-105A-C	11.0 kW (3x)	8.2 kW Running	415V / 3-Ph / 50Hz (IP55)
Packaging & Dedusting Line	PK-101	12.5 kW	9.4 kW Running	415V / 3-Ph / Cleanroom Duty
Total Connected Plant Load	MCC-01	184.5 kW	129.2 kW Operating	Diversity Factor: 0.70 (0.98 PF)
Substation Transformer	TR-01	500 kVA	131.8 kVA (26.4%)	11 kV / 433V DNAN Oil Immersed
Standby Diesel Generator	DG-01	250 kVA	200 kW Backup	AMF Auto Synchronized Panel
Evaporator Surface Condenser	HE-103	315.0 kWth	45.0 m3/h Water	Supply 32 C / Return 38 C
Instrument & Utility Air	COMP-01	15.0 kW	120 Nm3/h @ 7 bar	ISO 8573-1 Class 1.2.1 Oil-Free

STEAM SYSTEM: Condensate recovery 85% returned to deaerator tank @ 90 deg C via trap network.

ELECTRICAL SYSTEM: Central APFC automatic capacitor bank maintaining PF > 0.98 at 415V busbar.

COMPRESSED AIR: Refrigerated air dryer providing +3 deg C pressure dew point for pneumatic cylinders.

COOLING WATER: Induced draft FRP cooling tower (50 m3/h capacity) with delta T = 6.0 deg C.

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

PROCESS STREAM TABLE & MASS-ENERGY BALANCE
DISCIPLINE: PROCESS | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-002 | REV: V8.0 | SHEET: 2 OF 12

SCALE: AS SHOWN | DATE: 2026-10-17 | STATUS: PRELIMINARY | GATE: G-A HOLD

CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT | COPYRIGHT FINNOAQ